

ature, for example, gallium arsenide (GaAs) and indium arsenide (InAs). Spinning of electrons can be easily controlled and managed in semiconductors and such devices can be easily integrated with present semiconductor architecture and technology. Based upon spintronics we have devices like spin-transistors, spin-LEDs, optical switches, memory devices, and so on.

Because of unique science behind this technology, the operation of spintronics sensors and devices is more precise, reliable, and repeatable. Compared to electromechanical sensors, spintronics sensors are immune to mechanical shock, vibration, and temperature. Due to these very precise properties, the sensors are being used in Class I medical device hearing aids and providing miniaturisation, stability, and ease of operation with better sensitivity. Spintronics devices and sensors are also used in Class III devices like pacemakers and implants, where all the above properties are the key for their use.

Spintronics devices are being used to detect diseases in a human body, like cancers. The fact that the affected organ has different magnetic property than the normal one is being harnessed in this detection. And this is being used as a major game changer in the medical field. Electron transport in DNA is another field of great importance and research that is being carried out.

Spintronics devices are also used in mass storage. The storage density, which is the amount of data stored in a physical area of a chip, is much higher than conventional one. For example, it can store more than one trillion bits per square inch. Another game changer of the future may be the spin based quantum computation in nano technology, where the neuromorphic computing and quantum computing will take the front seat.

Molecular electronics

The goal of electronics is miniaturisation. Molecular electronics is that branch of electronics, or for that matter research, where a single molecule or a bunch of molecules is treated as electronics building blocks. Whereas our conventional electronics is based on the bulk of material, in single molecule electronics—that is, molecular electronics—this bulk of material is replaced with a single molecule.

Molecular electronics as a potential technology is basically a bottom-up approach where the smallest building unit, a molecule, is assembled to perform electronic operation. Whereas, the top-down approach is to shrink macroscopic components and systems.

The origin of molecular electronics can be traced back to 1974 when Mark Ratner and Arieh Aviram published the study of electron transport through a single molecule. Their suggestion of an ad-hoc calculation and the proposal that a single molecule could be used as a device has led to this branch of electronics.

Electron movement by a single molecule is at the centre of molecular electronics. This concept of electron movement through a single molecule is described in two different ways. One way is the movement of an electron through a single molecule, which is described as electron transfer. The other is the movement of current passing through a single molecule.

In molecular electronics, the goal of miniaturisation is achieved and due to this power consumption for its operation reduces drastically, whereas its sensitivity increases exponentially. Another major advantage of molecular electronics is that some of the molecular systems have the tendency to self-assemble to a bigger functional block. In this phenomenon, the components of a system come together spontaneously

due to some environmental factor to form a large functional unit.

The dimensionally small molecular circuit or electronic component has a wide range and variety of electrical, mechanical, optical, and thermal properties. And each of these properties in singular as well as in combined state gives rise to countless new physical phenomenon.

Molecular electronics, in principle, offers following advantages over the conventional electronics:

- The reduced size of each building block—the molecule—will offer higher packing density of the devices. This will subsequently reduce the cost and power and improve the efficiency and effectiveness.

- Molecular wire could reduce the transit time of a transistor, thereby reducing the time of operation.

- Molecular self-assembly can be used to modify the electronic behaviour of an electronic component to provide both switching and sensing capability on a molecular scale.

- Special properties of a molecule, such as stable geometric structure, could lead to a new dimension in electronics, which is otherwise not possible in conventional electronics.

- By intelligent choice of composition and geometry, one can create variation in molecular transport, binding, optical, and structural property.

But molecules have the obvious disadvantage of being unstable at high temperatures. Moreover, fabrication of a reliable molecular junction requires control of unprecedented level. Achieving the same is not only difficult but slow and costly too.

Several molecular electronic solutions have already been developed, such as molecular wire, single-molecule transistor, rectifiers, and so on. However, molecular electronics is still in infant stage of research and these devices are yet to come out of laboratory bed. **EFY**